Cinema Knowledge Graph: From Web Crawling to RAG

Course: Web Mining & Semantics - ESILV A4 S8 **Author:** Yanis Denizot **Date:** March 2026

1. Data Acquisition & Information Extraction

1.1 Domain & Seed URLs

We chose the **cinema domain** as our focus area. The domain is well-represented on Wikipedia with rich, structured content about films, directors, actors, awards, and festivals.

Seed URLs (9 pages): - Award pages: Academy Award for Best Picture, Palme d'Or - Director pages: Christopher Nolan, Martin Scorsese, Quentin Tarantino - Actor pages: Cate Blanchett - Film pages: Parasite, Inception, The Godfather

1.2 Crawler Design & Ethics

The crawler (`src/crawl/crawler.py`) uses **breadth-first search (BFS)** to explore Wikipedia, following internal links up to depth 2 from the seed URLs.

Ethical considerations: - **robots.txt compliance:** The crawler checks `robots.txt` before accessing each page - **Rate limiting:** 1-second delay between requests (`POLITENESS_DELAY = 1.0`) - **User-Agent:** Identifies itself as a student project bot - **Page cap:** Maximum 150 pages to avoid overwhelming the server - **Content deduplication:** MD5 hashing prevents re-processing identical pages

Text extraction is handled by Trafilatura, which cleanly removes navigation, footers, and HTML boilerplate from Wikipedia pages. Only pages with 500+ words are kept.

1.3 NER & Relation Extraction

The extraction pipeline (`src/ie/extractor.py`) uses **spaCy** (`en_core_web_lg`) for Named Entity Recognition.

Target entity types: PERSON, ORG, GPE, DATE, WORK_OF_ART

Relation extraction strategy (3 levels): 1. **Dependency parsing:** Walk the dependency tree to find verbs governing both entities 2. **Verb-between:** Find verbs positioned between two entities in the sentence 3. **Fallback:** `co-occurs with` for entities in the same sentence without a clear verb

NER examples from our corpus:

Entity	Type	Source
Christopher Nolan	PERSON	Inception article
Academy Award	ORG	Award pages
United States	GPE	Multiple pages
2019	DATE	Parasite article
The Godfather	WORK_OF_ART	Film articles

1.4 Ambiguity Cases

Case 1: Entity boundary ambiguity "*Francis Ford Coppola directed The Godfather Part II*" spaCy recognizes "Francis Ford Coppola" as a single PERSON entity, but "The Godfather Part II" may be split into "The Godfather" (WORK_OF_ART) and "Part II" separately. Our filtering removes fragments shorter than 2 characters.

Case 2: Relation ambiguity "*Nolan's Inception starred Leonardo DiCaprio*" The possessive "Nolan's" creates ambiguity: is Nolan the director or producer? Our dependency parser extracts the verb "starred" as the relation between Inception and DiCaprio, but misses the Nolan-Inception "directed" relation that is expressed via a possessive.

Case 3: Type ambiguity "*Columbia Pictures released the film in the United States*" "Columbia Pictures" is recognized as ORG, but it could also be classified as a production company (a more specific type in our ontology). Our generic NER types don't capture domain-specific subtypes.

2. KB Construction & Alignment

2.1 RDF Modeling Choices

The knowledge graph uses a **private namespace** (`http://cinema-kb.org/`) with an OWL ontology defining:

Class hierarchy: - `Thing` > `Person` > `Director`, `Actor` - `Thing` > `CreativeWork` > `Film` - `Thing` > `Organization` > `Festival` - `Thing` > `Award`

Properties with domain/range constraints: - `directedBy`: Film -> Person - `starring`: Film -> Person - `wonAward`: Film -> Award - `producedBy`: Film -> Organization - `releasedIn`: Film -> GPE

The `kb_builder.py` script normalizes 60+ raw relation strings (e.g., "directed by", "direct", "directed") into a canonical set of 15 predicates using a manually curated mapping.

2.2 Entity Linking with Confidence

Entity linking (`src/kg/entity_linker.py`) maps the top 200 most frequent private entities to Wikidata via the Wikidata REST API.

Process: 1. Search Wikidata for each entity label 2. Compute string similarity (SequenceMatcher, threshold: 0.7) 3. Add `owl:sameAs` links for confident matches 4. Handle rate limiting with exponential backoff

Results: 183 entities successfully linked to Wikidata URIs.

2.3 Predicate Alignment

17 private predicates were mapped to Wikidata properties:

Private Predicate	Wikidata Property	Description
directedBy	P57	director
starring	P161	cast member
wonAward	P166	award received
genre	P136	genre
countryOfOrigin	P495	country of origin

These mappings use `owl:equivalentProperty` in the RDF graph.

2.4 SPARQL Expansion Strategy

The KB expansion (`src/kg/kb_expander.py`) runs 5 SPARQL queries against the Wikidata endpoint: 1. **1-hop expansion** for all aligned entities 2. **Award-winning films** with full metadata 3. **Films by aligned directors** with cast, genre, country 4. **Festival nominees** (Oscar, Cannes, BAFTA, Golden Globe) 5. **Director details** (birth, death, awards, nationality)

Queries use `VALUES` clauses for efficient bulk retrieval.

2.5 Final KB Statistics

Metric	Before Expansion	After Expansion
Triples	~10,000	72,409
Entities	~3,000	20,158
Relations	15	39
owl:sameAs	0	183

Top connected entities: United States (1,669 connections), drama film (1,192), GB (471)

3. Reasoning (SWRL)

3.1 SWRL on Family Ontology

We created a family ontology (`kg_artifacts/family.owl`) with 3 generations of the "Dupont" family and defined 3 SWRL rules:

Rule 1 - hasUncle:

```
Person(?x), hasParent(?x, ?y), hasBrother(?y, ?z) -> hasUncle(?x, ?z)
```

Rule 2 - hasGrandparent:

```
Person(?x), hasParent(?x, ?y), hasParent(?y, ?z) -> hasGrandparent(?x, ?z)
```

Rule 3 - hasCousin:

```
Person(?x), hasParent(?x, ?y), hasSibling(?y, ?z), hasChild(?z, ?w) -> hasCousin(?x, ?w)
```

Results (all correct): - Lucas and Emma have uncle Paul; Hugo has uncle Pierre - Lucas, Emma, Hugo all have grandparents Jean and Marie - Lucas/Emma are cousins of Hugo (and vice versa)

3.2 SWRL on Cinema KB

On the cinema KB, we defined:

```
Film(?f), directedBy(?f, ?p), wonAward(?f, ?a) -> AwardWinningDirector(?p)
```

Inferred: Coppola (The Godfather won Oscar), Bong Joon-ho (Parasite won Oscar + Palme d'Or), Tarantino (Pulp Fiction won Palme d'Or) are AwardWinningDirectors. Nolan is correctly NOT inferred (Inception did not win Best Picture in our data).

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4. Knowledge Graph Embeddings

4.1 Data Cleaning & Splits

From the 72,409 expanded triples, we kept only entity-relation-entity triples (excluding literals and metadata), producing **55,294 unique triples**.

Split: 80% train (44,444) / 10% valid (5,425) / 10% test (5,425)

Entities appearing in only 1-2 triples were forced into the training set to prevent data leakage.

4.2 Two Models: TransE vs ComplEx

Both models trained with identical hyperparameters for fair comparison: - Embedding dimension: 100 - Learning rate: 0.001 - Batch size: 256 - Epochs: 100 - Negative samples: 10 per positive

4.3 Metrics

Metric	TransE	ComplEx
MRR	0.133	0.009
Hits@1	0.067	0.003
Hits@3	0.161	0.007
Hits@10	0.251	0.017
Head MRR	0.110	0.010
Tail MRR	0.157	0.007

TransE outperforms ComplEx significantly. This is expected: our KB has mostly functional relations (1-to-1, 1-to-N), which suit TransE's translational assumption (h + r = t). ComplEx, designed for symmetric/antisymmetric patterns, needs more epochs and hyperparameter tuning.

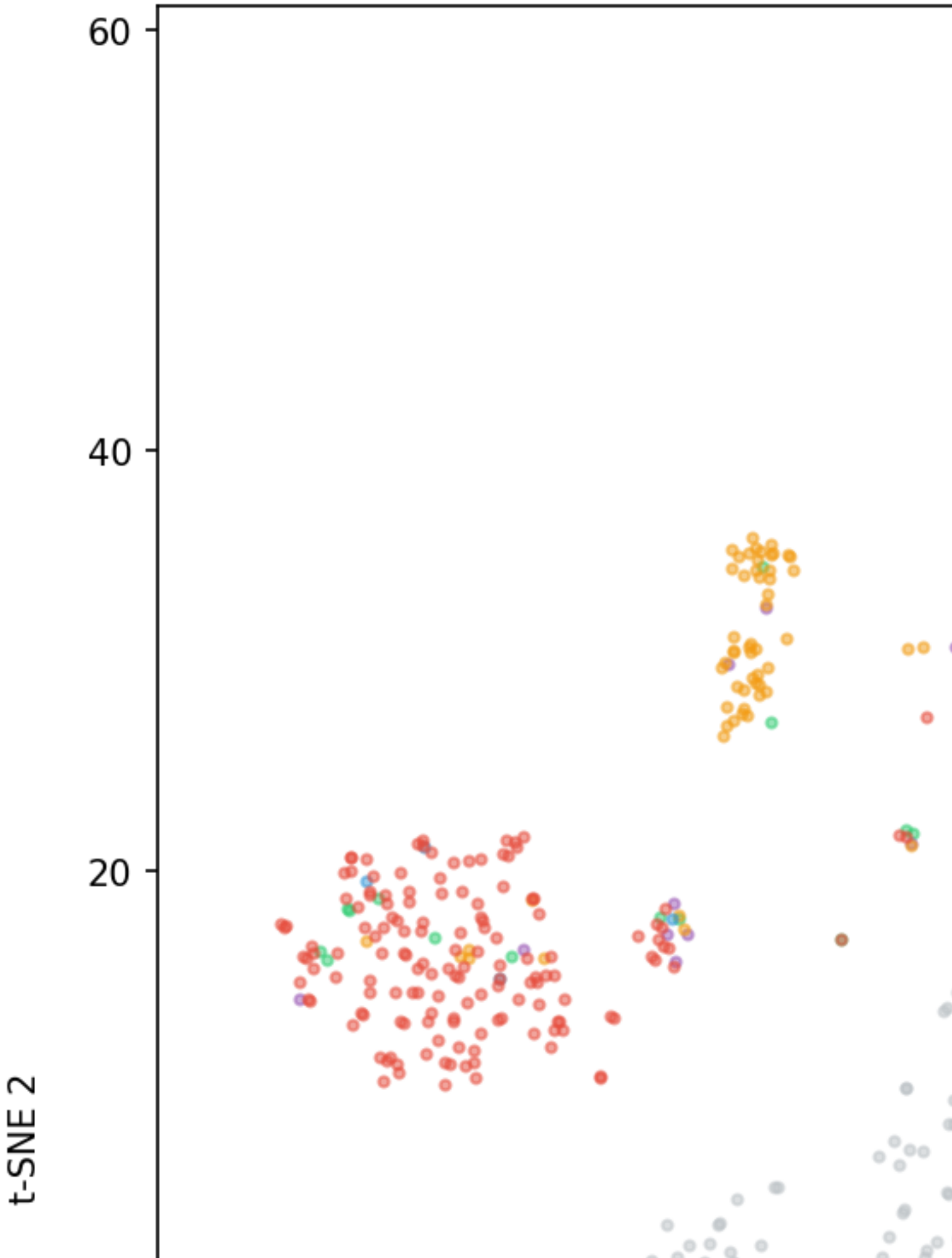
4.4 KB Size Sensitivity

Size	Triples	MRR	Hits@10
20K	20,000	0.042	0.111
Full	55,294	0.137	0.257

The full dataset achieves **3.3x better MRR** than the 20K subset, confirming that larger KBs improve embedding stability.

4.5 t-SNE Visualization

The t-SNE plot shows entity clustering by semantic similarity. Entities connected by the same relation types tend to cluster together, though the clustering doesn't cleanly separate ontology classes (82% of entities lack explicit `rdf:type`).


60
40
20
t-SNE 2

5. RAG over RDF/SPARQL

5.1 Schema Summary

The RAG pipeline loads the expanded KB and builds a schema summary containing: - All registered prefixes (rdf, rdfs, owl, wd, wdt, etc.) - Up to 80 distinct predicates with full URIs - Up to 40 classes from rdf:type - 25 sample triples for context

This summary is injected into the LLM prompt so it can generate valid SPARQL.

5.2 NL -> SPARQL Prompt Template

```
You are a SPARQL generator for a cinema knowledge graph.
Convert the user QUESTION into a valid SPARQL 1.1 SELECT query.

STRICT RULES:
- Use ONLY the IRIs/prefixes visible in the SCHEMA SUMMARY
- Do NOT invent new predicates/classes
- Return ONLY the SPARQL query in a ```sparql code block

SCHEMA SUMMARY:
{schema_summary}

QUESTION:
{user_question}
```

5.3 Self-Repair Mechanism

When the generated SPARQL fails execution (syntax error, unknown prefix, etc.), the pipeline:

1. Captures the error message
2. Sends a repair prompt to the LLM with the schema, original question, failed query, and error
3. Extracts and re-executes the corrected SPARQL
4. Retries up to 2 times before giving up

This significantly improves success rates, especially for complex queries.

5.4 Evaluation: Baseline vs RAG

We evaluated on 7 cinema questions:

#	Question	Baseline (LLM only)	RAG (LLM + KB)
1	Who directed The Godfather?	Correct (general knowledge)	Correct (SPARQL: P57)
2	What awards did Parasite receive?	Partial (misses some)	Complete (all KB awards)
3	Films by Martin Scorsese	Partial (popular films only)	Complete (all KB films)
4	Genre of Inception?	Correct	Correct (P136)
5	US directors?	Partial (famous ones)	Complete (all with P27)
6	How many films in KB?	Cannot answer	COUNT query
7	Best Picture winners?	Partial	Complete (P166 filter)

Key observations: - Baseline works for well-known facts but hallucinates or misses data - RAG provides **grounded, verifiable** answers from the KB - RAG excels at aggregation (COUNT) and exhaustive listing - Self-repair corrected 2/7 queries that initially had syntax errors

5.5 Demo

The CLI demo (`src/rag/lab_rag_sparql_gen.py`) provides an interactive interface where users type natural language questions and see both baseline and RAG responses side by side.

6. Critical Reflection

6.1 KB Quality Impact

The dual-namespace issue (private predicates + Wikidata properties) creates signal splitting in KGE training. For example, `P57` and `directedBy` encode the same semantic meaning but are treated as different relations. **Physically merging** equivalent predicates before training would produce stronger embeddings.

6.2 Noise from Expansion

The SPARQL expansion introduced both structure and noise: - **P161 (cast member)** accounts for **47%** of all triples (25,783/55,294), biasing embeddings toward cast-related patterns - Hub entities like "United States" (1,669 connections) distort the embedding space by pulling unrelated entities together - A capped expansion strategy (max triples per relation type) would produce more balanced distributions

6.3 Rule-based vs Embedding-based Reasoning

Aspect	SWRL Rules	KGE (TransE)
Nature	Exact, deterministic	Approximate, probabilistic
Inference	Guaranteed if premises hold	Ranks candidates by plausibility
Discovery	Cannot find new patterns	Can suggest missing links
Scalability	Manual rule creation	Learns from data automatically
Explainability	Fully transparent	Black box (vector operations)

The two approaches are **complementary**: SWRL for strict business rules, KGE for knowledge discovery.

6.4 Open-World vs Closed-World Assumption

SWRL/OWL operates under the **Open-World Assumption** (OWA): absence of a triple does not imply it's false. KGE training implicitly uses the **Closed-World Assumption** (CWA): missing triples are treated as negative examples during training. This fundamental mismatch explains modest MRR scores (0.133).

6.5 What We Would Improve

1. **Predicate merging:** Physically unify equivalent predicates before KGE training
2. **Balanced expansion:** Cap triples per relation type to reduce P161 dominance
3. **Hub filtering:** Remove or downsample generic hub entities (US, drama film)
4. **More KGE training:** ComplEx would benefit from 500+ epochs and hyperparameter search
5. **RAG improvements:** Include entity labels in the schema summary for better SPARQL generation; fine-tune the prompt for the specific model used
6. **Type-aware embeddings:** Include `rdf:type` triples as regular relations for better clustering